

Red Hat OpenShift Data Foundation 4.22

Replacing devices

Instructions for safely replacing operational or failed devices

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Abstract

This document explains how to safely replace storage devices for Red Hat OpenShift Data Foundation.

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PREFACE

This document describes how to replace operational and failed storage devices in your OpenShift Data Foundation cluster.

PROVIDING FEEDBACK ON RED HAT DOCUMENTATION

We appreciate your input on our documentation. Do let us know how we can make it better.

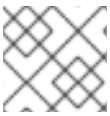
To give feedback, create a Jira ticket:

Procedure

1. Log in to the Jira.
2. Click **Create** in the top navigation bar
3. Enter a descriptive title in the Summary field.
4. Enter your suggestion for improvement in the Description field. Include links to the relevant parts of the documentation.
5. Select **Documentation** in the Components field.
6. Click Create at the bottom of the dialogue.

CHAPTER 1. REPLACING STORAGE DEVICES OVERVIEW

Depending on the type of your deployment, you can choose one of the following procedures to replace a storage device:



NOTE

OpenShift Data Foundation does not support heterogeneous OSD sizes.

- Replacing operational or failed storage devices on AWS user-provisioned infrastructure
- Replacing operational or failed storage devices on AWS installer-provisioned infrastructure
- Replacing operational or failed storage devices on VMware infrastructure
- Replacing operational or failed storage devices on Azure installer-provisioned infrastructure
- Replacing operational or failed storage devices on clusters backed by local storage devices
- Replacing operational or failed storage devices on IBM Power

Additional resources

- Section 2.1, “Replacing operational or failed storage devices on AWS user-provisioned infrastructure”
- Section 2.2, “Replacing operational or failed storage devices on AWS installer-provisioned infrastructure”
- Section 3.1, “Replacing operational or failed storage devices on VMware infrastructure”
- Section 4.1, “Replacing operational or failed storage devices on Azure installer-provisioned infrastructure”
- Section 6.1, “Replacing operational or failed storage devices on clusters backed by local storage devices”
- Section 6.3, “Replacing operational or failed storage devices on IBM Power”

CHAPTER 2. REPLACING DYNAMICALLY PROVISIONED STORAGE DEVICES ON AWS

2.1. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON AWS USER-PROVISIONED INFRASTRUCTURE

When you need to replace a device in a dynamically created storage cluster on an AWS user-provisioned infrastructure, you must replace the storage node.

Additional resources

- Replacing an operational AWS node on user-provisioned infrastructure
- Replacing a failed AWS node on user-provisioned infrastructure

2.2. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON AWS INSTALLER-PROVISIONED INFRASTRUCTURE

When you need to replace a device in a dynamically created storage cluster on an AWS installer-provisioned infrastructure, you must replace the storage node.

Additional resources

- Replacing an operational AWS node on installer-provisioned infrastructure
- Replacing a failed AWS node on installer-provisioned infrastructure

CHAPTER 3. REPLACING DYNAMICALLY PROVISIONED STORAGE DEVICES ON VMWARE

3.1. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON VMWARE INFRASTRUCTURE

Create a new Persistent Volume Claim (PVC) on a new volume, and remove the old object storage device (OSD) when one or more virtual machine disks (VMDK) needs to be replaced in OpenShift Data Foundation which is deployed dynamically on VMware infrastructure.

Prerequisites

- Ensure that the data is resilient.
  - In the OpenShift Web Console, click **Storage → Data Foundation**.
  - Click the **Storage Systems** tab, and then click **ocs-storagecluster**.
  - In the **Status card** of **Block and File** dashboard, under the **Overview** tab, verify that *Data Resiliency* has a green tick mark.

Procedure

1. Identify the OSD that needs to be replaced and the OpenShift Container Platform node that has the OSD scheduled on it.

```
$ oc get -n openshift-storage pods -l app=rook-ceph-osd -o wide
```

   Example output:

```
rook-ceph-osd-0-6d77d6c7c6-m8xj6   0/1   CrashLoopBackOff   0   24h   10.129.0.16
compute-2   <none>          <none>
rook-ceph-osd-1-85d99fb95f-2svc7   1/1   Running            0   24h   10.128.2.24   compute-
0   <none>          <none>
rook-ceph-osd-2-6c66cdb977-jp542   1/1   Running            0   24h   10.130.0.18   compute-
1   <none>          <none>
```

   In this example, **rook-ceph-osd-0-6d77d6c7c6-m8xj6** needs to be replaced and **compute-2** is the OpenShift Container platform node on which the OSD is scheduled.

   NOTE

   The status of the pod is **Running**, if the OSD you want to replace is healthy.

2. Scale down the **rook-ceph-operator** and **ocs-operator** deployments.

```
$ oc -n openshift-storage scale deployment rook-ceph-operator --replicas=0
```

```
$ oc -n openshift-storage scale deployment ocs-operator --replicas=0
```

3. Scale down the OSD deployment for the OSD to be replaced.

Each time you want to replace the OSD, update the **osd_id_to_remove** parameter with the OSD ID, and repeat this step.

```
$ osd_id_to_remove=0
```

```
$ oc scale -n openshift-storage deployment rook-ceph-osd-${osd_id_to_remove} --replicas=0
```

where, **osd_id_to_remove** is the integer in the pod name immediately after the **rook-ceph-osd** prefix. In this example, the deployment name is **rook-ceph-osd-0**.

Example output:

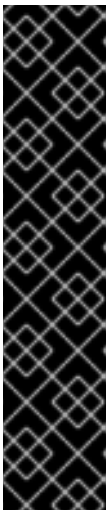
```
deployment.extensions/rook-ceph-osd-0 scaled
```

4. Verify that the **rook-ceph-osd** pod is terminated.

```
$ oc get -n openshift-storage pods -l ceph-osd-id=${osd_id_to_remove}
```

Example output:

```
No resources found.
```



IMPORTANT

If the **rook-ceph-osd** pod is in **terminating** state, use the **force** option to delete the pod.

```
$ oc delete pod rook-ceph-osd-0-6d77d6c7c6-m8xj6 --force --grace-period=0
```

Example output:

```
warning: Immediate deletion does not wait for confirmation that the running
resource has been terminated. The resource may continue to run on the
cluster indefinitely.
  pod "rook-ceph-osd-0-6d77d6c7c6-m8xj6" force deleted
```

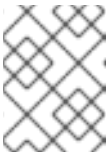
5. Remove the old OSD from the cluster so that you can add a new OSD.

   a. Delete any old **ocs-osd-removal** jobs.

```
$ oc delete -n openshift-storage job ocs-osd-removal-job
```

   Example output:

```
job.batch "ocs-osd-removal-job" deleted
```



NOTE

If the above job does not reach **Completed** state after 10 minutes, then the job must be deleted and rerun with **FORCE_OSD_REMOVAL=true**.

b. Navigate to the **openshift-storage** project.

```
$ oc project openshift-storage
```

c. Remove the old OSD from the cluster.

```
$ oc process -n openshift-storage ocs-osd-removal -p
FAILED_OSD_IDS=${osd_id_to_remove} FORCE_OSD_REMOVAL=false |oc create -n
openshift-storage -f -
```

Set **FORCE_OSD_REMOVAL** to **true** when removing an OSD leaves the cluster with insufficient OSDs to satisfy its data protection requirements. Without this parameter, the Ceph safe-to-destroy check does not pass and the removal job hangs indefinitely.



WARNING

Proceeding will permanently remove the OSD from the cluster and destroy its data. Ensure **osd_id_to_remove** is correct. Do not process multiple IDs from the same failure domain at once. Contact Red Hat Technical Support if you need assistance.

6. Verify that the OSD was removed successfully by checking the status of the **ocs-osd-removal-job** pod.
A status of **Completed** confirms that the OSD removal job succeeded.

```
$ oc get pod -l job-name=ocs-osd-removal-job -n openshift-storage
```

7. Ensure that the OSD removal is completed.

```
$ oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1 | egrep -i 'completed
removal'
```

Example output:

```
2022-05-10 06:50:04.501511 I | cephosd: completed removal of OSD 0
```



IMPORTANT

If the **ocs-osd-removal-job** pod fails and the pod is not in the expected **Completed** state, check the pod logs for further debugging.

For example:

```
# oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1
```

8. If encryption was enabled at the time of install, remove **dm-crypt** managed **device-mapper** mapping from the OSD devices that are removed from the respective OpenShift Data Foundation nodes.

   a. Get the PVC name(s) of the replaced OSD(s) from the logs of **ocs-osd-removal-job** pod.

   ```
   $ oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1  |egrep -i ‘pvc|deviceset’
   ```

   Example output:

   ```
   2021-05-12 14:31:34.666000 I | cephosd: removing the OSD PVC "ocs-deviceset-xxxx-xxx-xxx-xxx"
   ```

   b. For each of the previously identified nodes, do the following:

      i. Create a **debug** pod and **chroot** to the host on the storage node.

      ```
      $ oc debug node/<node name>
      ```

      <node name>

      Is the name of the node.

      ```
      $ chroot /host
      ```

      ii. Find a relevant device name based on the PVC names identified in the previous step.

      ```
      $ dmsetup ls| grep <pvc name>
      ```

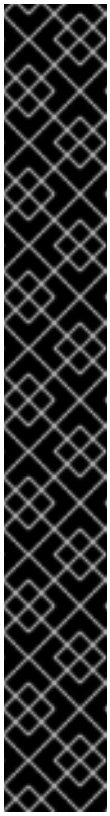
      <pvc name>

      Is the name of the PVC.
      Example output:

      ```
      ocs-deviceset-xxx-xxx-xxx-xxx-block-dmcrypt (253:0)
      ```

      iii. Remove the mapped device.

      ```
      $ cryptsetup luksClose --debug --verbose ocs-deviceset-xxx-xxx-xxx-xxx-block-dmcrypt
      ```



IMPORTANT

If the above command gets stuck due to insufficient privileges, run the following commands:

- Press **CTRL+Z** to exit the above command.
- Find the PID of the process which was stuck.

```
$ ps -ef | grep crypt
```

- Terminate the process using the **kill** command.

```
$ kill -9 <PID>
```

<PID>

Is the process ID.

- Verify that the device name is removed.

```
$ dmsetup ls
```

9. Delete the **ocs-osd-removal** job.

```
$ oc delete -n openshift-storage job ocs-osd-removal-job
```

Example output:

```
job.batch "ocs-osd-removal-job" deleted
```



NOTE

When using an external key management system (KMS) with data encryption, the old OSD encryption key can be removed from the Vault server as it is now an orphan key.

10. Scale up the **rook-ceph-operator** and **ocs-operator** deployments.

```
$ oc -n openshift-storage scale deployment rook-ceph-operator --replicas=1
```

```
$ oc -n openshift-storage scale deployment ocs-operator --replicas=1
```

Verification

1. Verify that there is a new OSD running.

```
$ oc get -n openshift-storage pods -l app=rook-ceph-osd
```

Example output:

```
rook-ceph-osd-0-5f7f4747d4-snshw                 1/1     Running     0          4m47s
rook-ceph-osd-1-85d99fb95f-2svc7                 1/1     Running     0          1d20h
rook-ceph-osd-2-6c66cdb977-jp542                 1/1     Running     0          1d20h
```

2. Verify that there is a new PVC created which is in **Bound** state.

```
$ oc get -n openshift-storage pvc
```

Example output:

```
NAME                      STATUS   VOLUME                                     CAPACITY   ACCESS
MODES   STORAGECLASS   AGE
ocs-deviceset-0-0-2s6w4   Bound    pvc-7c9bcaf7-de68-40e1-95f9-0b0d7c0ae2fc   512Gi
RWO           thin           5m
ocs-deviceset-1-0-q8fwh   Bound    pvc-9e7e00cb-6b33-402e-9dc5-b8df4fd9010f   512Gi
RWO           thin           1d20h
ocs-deviceset-2-0-9v8lq   Bound    pvc-38cdfcee-ea7e-42a5-a6e1-aaa6d4924291   512Gi
RWO           thin           1d20h
```

3. Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.

   a. Identify the nodes where the new OSD pods are running.

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
```

<OSD-pod-name>

Is the name of the OSD pod.
For example:

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
```

Example output:

```
NODE
compute-1
```

   b. For each of the nodes identified in the previous step, do the following:

      i. Create a debug pod and open a chroot environment for the selected host(s).

```
$ oc debug node/<node name>
```

<node name>

Is the name of the node.

```
$ chroot /host
```

      ii. Check for the **crypt** keyword beside the **ocs-deviceset** name(s).

```
$ lsblk
```

4. Log in to OpenShift Web Console and view the storage dashboard.

CHAPTER 4. REPLACING DYNAMICALLY PROVISIONED STORAGE DEVICES ON MICROSOFT AZURE

4.1. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON AZURE INSTALLER-PROVISIONED INFRASTRUCTURE

When you need to replace a device in a dynamically created storage cluster on an Azure installer-provisioned infrastructure, you must replace the storage node.

Additional resources

- Replacing operational nodes on Azure installer-provisioned infrastructure
- Replacing failed nodes on Azure installer-provisioned infrastructures

CHAPTER 5. REPLACING DYNAMICALLY PROVISIONED STORAGE DEVICES ON GOOGLE CLOUD

5.1. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON GOOGLE CLOUD INSTALLER-PROVISIONED INFRASTRUCTURE

When you need to replace a device in a dynamically created storage cluster on an Google Cloud installer-provisioned infrastructure, you must replace the storage node.

Additional resources

- Replacing operational nodes on Google Cloud installer-provisioned infrastructure
- Replacing failed nodes on Google Cloud installer-provisioned infrastructures

CHAPTER 6. REPLACING STORAGE DEVICES ON CLUSTERS DEPLOYED USING LOCAL STORAGE DEVICES

6.1. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON CLUSTERS BACKED BY LOCAL STORAGE DEVICES

You can replace an object storage device (OSD) in OpenShift Data Foundation deployed using local storage devices on the following infrastructures:

- Bare metal / Agnostic deployment
- VMware
- IBM Z and IBM® LinuxONE



NOTE

There might be a need to replace one or more underlying storage devices.

Prerequisites

- Red Hat recommends that replacement devices are configured with similar infrastructure and resources to the device being replaced.
- Ensure that the data is resilient.
  - In the OpenShift Web Console, click **Storage → Data Foundation**.
  - Click the **Storage Systems** tab, and then click **ocs-storagecluster**.
  - In the **Status card** of **Block and File** dashboard, under the **Overview** tab, verify that *Data Resiliency* has a green tick mark.

Procedure

1. Remove the underlying storage device from relevant worker node.
2. Verify that relevant OSD Pod has moved to CrashLoopBackOff state.
   Identify the OSD that needs to be replaced and the OpenShift Container Platform node that has the OSD scheduled on it.

   ```
   $ oc get -n openshift-storage pods -l app=rook-ceph-osd -o wide
   ```

   Example output:

   ```
   rook-ceph-osd-0-6d77d6c7c6-m8xj6   0/1   CrashLoopBackOff   0   24h   10.129.0.16
   compute-2   <none>          <none>
   rook-ceph-osd-1-85d99fb95f-2svc7   1/1   Running            0   24h   10.128.2.24   compute-
   0   <none>          <none>
   rook-ceph-osd-2-6c66cdb977-jp542   1/1   Running            0   24h   10.130.0.18   compute-
   1   <none>          <none>
   ```

In this example, **rook-ceph-osd-0-6d77d6c7c6-m8xj6** needs to be replaced and **compute-2** is the OpenShift Container platform node on which the OSD is scheduled.

3. Scale down the **rook-ceph-operator** and **ocs-operator** deployments.

```
$ oc -n openshift-storage scale deployment rook-ceph-operator --replicas=0
```

```
$ oc -n openshift-storage scale deployment ocs-operator --replicas=0
```

4. Scale down the OSD deployment for the OSD to be replaced.

```
$ osd_id_to_remove=0
```

```
$ oc scale -n openshift-storage deployment rook-ceph-osd-${osd_id_to_remove} --replicas=0
```

where, **osd_id_to_remove** is the integer in the pod name immediately after the **rook-ceph-osd** prefix. In this example, the deployment name is **rook-ceph-osd-0**.

Example output:

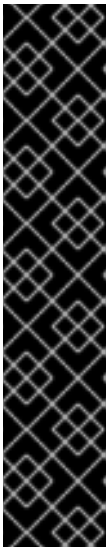
```
deployment.extensions/rook-ceph-osd-0 scaled
```

5. Verify that the **rook-ceph-osd** pod is terminated.

```
$ oc get -n openshift-storage pods -l ceph-osd-id=${osd_id_to_remove}
```

Example output:

```
No resources found in openshift-storage namespace.
```



IMPORTANT

If the **rook-ceph-osd** pod is in **terminating** state for more than a few minutes, use the **force** option to delete the pod.

```
$ oc delete -n openshift-storage pod rook-ceph-osd-0-6d77d6c7c6-m8xj6 --grace-period=0 --force
```

Example output:

```
warning: Immediate deletion does not wait for confirmation that the running resource has been terminated. The resource may continue to run on the cluster indefinitely.
 pod "rook-ceph-osd-0-6d77d6c7c6-m8xj6" force deleted
```

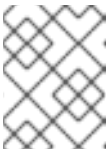
6. Remove the old OSD from the cluster so that you can add a new OSD.

   a. Delete any old **ocs-osd-removal** jobs.

```
$ oc delete -n openshift-storage job ocs-osd-removal-job
```

Example output:

```
job.batch "ocs-osd-removal-job" deleted
```



NOTE

The above command must reach **Completed** state before moving to the next steps. This can take more than ten minutes.

b. Navigate to the **openshift-storage** project.

```
$ oc project openshift-storage
```

c. Remove the old OSD from the cluster.

```
$ oc process -n openshift-storage ocs-osd-removal -p
FAILED_OSD_IDS=${osd_id_to_remove} FORCE_OSD_REMOVAL=false |oc create -n
openshift-storage -f -
```

Set **FORCE_OSD_REMOVAL** to **true** when removing an OSD leaves the cluster with insufficient OSDs to satisfy its data protection requirements. Without this parameter, the Ceph safe-to-destroy check does not pass and the removal job hangs indefinitely.



WARNING

Proceeding will permanently remove the OSD from the cluster and destroy its data. Ensure **osd_id_to_remove** is correct. Do not process multiple IDs from the same failure domain at once. Contact Red Hat Technical Support if you need assistance.

7. Verify that the OSD was removed successfully by checking the status of the **ocs-osd-removal-job** pod.
   A status of **Completed** confirms that the OSD removal job succeeded.

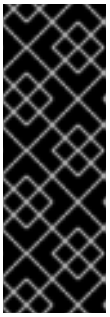
```
$ oc get pod -l job-name=ocs-osd-removal-job -n openshift-storage
```

8. Ensure that the OSD removal is completed.

```
$ oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1 | egrep -i 'completed
removal'
```

Example output:

```
2022-05-10 06:50:04.501511 I | cephosd: completed removal of OSD 0
```



IMPORTANT

If the **ocs-osd-removal-job** fails and the pod is not in the expected **Completed** state, check the pod logs for further debugging.

For example:

```
# oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1
```

9. If encryption was enabled at the time of install, remove **dm-crypt** managed **device-mapper** mapping from the OSDs that are removed from the respective OpenShift Data Foundation nodes.

   a. Get the Persistent Volume Claim (PVC) names of the replaced OSDs from the logs of **ocs-osd-removal-job** pod.

   ```
   $ oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1  |egrep -i 'pvc|deviceset'
   ```

   Example output:

   ```
   2021-05-12 14:31:34.666000 I | cephosd: removing the OSD PVC "ocs-deviceset-xxxx-xxx-xxx-xxx"
   ```

   b. For each of the previously identified nodes, do the following:

      i. Create a **debug** pod and **chroot** to the host on the storage node.

      ```
      $ oc debug node/<node name>
      ```

      <node name>

      Is the name of the node.

      ```
      $ chroot /host
      ```

      ii. Find the relevant device name based on the PVC names identified in the previous step.

      ```
      $ dmsetup ls| grep <pvc name>
      ```

      <pvc name>

      Is the name of the PVC.
      Example output:

      ```
      ocs-deviceset-xxx-xxx-xxx-xxx-block-dmcrypt (253:0)
      ```

      iii. Remove the mapped device.

      ```
      $ cryptsetup luksClose --debug --verbose <ocs-deviceset-name>
      ```

      <ocs-deviceset-name>

Is the name of the relevant device based on the PVC names identified in the previous step.



IMPORTANT

If the above command gets stuck due to insufficient privileges, run the following commands:

- Press **CTRL+Z** to exit the above command.
- Find the PID of the process which was stuck.

```
$ ps -ef | grep crypt
```

- Terminate the process using the **kill** command.

```
$ kill -9 <PID>
```

<PID>

Is the process ID.

- Verify that the device name is removed.

```
$ dmsetup ls
```

10. Find the persistent volume (PV) that need to be deleted.

```
$ oc get pv -L kubernetes.io/hostname | grep <storageclass-name> | grep Released
```

Example output:

```
local-pv-d6bf175b 1490Gi RWO Delete Released openshift-storage/ocs-deviceset-0-data-0-6c5pw localblock 2d22h compute-1
```

11. Delete the PV.

```
$ oc delete pv <pv_name>
```

12. Physically add a new device to the node.

13. Track the provisioning of PVs for the devices that match the **deviceInclusionSpec**. It can take a few minutes to provision the PVs.

```
$ oc -n openshift-local-storage describe localvolumeset <lvs-name>
```

Example output:

```
[...]
Status:
  Conditions:
    Last Transition Time:         2020-11-17T05:03:32Z
```

```
    Message:                DiskMaker: Available, LocalProvisioner: Available
    Status:                 True
    Type:                   DaemonSetsAvailable
    Last Transition Time:       2020-11-17T05:03:34Z
    Message:                Operator reconciled successfully.
    Status:                 True
    Type:                   Available
  Observed Generation:          1
  Total Provisioned Device Count: 4
Events:
Type    Reason      Age         From              Message
----    ------      ----        ----              -------
Normal  Discovered  2m30s (x4   localvolumeset-    node.example.com -
        NewDevice   over 2m30s) symlink-controller found possible
                                matching disk,
                                waiting 1m to claim

Normal  FoundMatch  89s (x4     localvolumeset-    node.example.com -
        ingDisk     over 89s)   symlink-controller symlinking matching
                                disk
```

Once the PV is provisioned, a new OSD pod is automatically created for the PV.

14. Delete the **ocs-osd-removal** job(s).

```
$ oc delete -n openshift-storage job ocs-osd-removal-job
```

Example output:

```
job.batch "ocs-osd-removal-job" deleted
```



NOTE

When using an external key management system (KMS) with data encryption, the old OSD encryption key can be removed from the Vault server as it is now an orphan key.

15. Scale up the **rook-ceph-operator** and **ocs-operator** deployments.

```
$ oc -n openshift-storage scale deployment rook-ceph-operator --replicas=1
```

```
$ oc -n openshift-storage scale deployment ocs-operator --replicas=1
```

16. [Optional] Verify the Ceph cluster health and check if there are any crash reports associated with the previous failed OSD which has now been replaced.

    a. List all crash reports:

```
$ oc exec -it $(oc get pod -n openshift-storage -l app=rook-ceph-operator -o name) -n
openshift-storage -- ceph crash ls -c /var/lib/rook/openshift-storage/openshift-
storage.config
```

    Sample output:

```
ID                              ENTITY  NEW
2025-09-25T07:12:45Z_1234567890   osd.5   *
2025-09-25T08:20:11Z_abcdef1234   mon.a   *
```

b. Filter crashes for the failed OSD from the previous step. For example, if the failed OSD is **osd.5**, run:

```
$ oc exec -it $(oc get pod -n openshift-storage -l app=rook-ceph-operator -o name) -n openshift-storage -- ceph crash ls -c /var/lib/rook/openshift-storage/openshift-storage.config | grep osd.5
```

Note the specif crash report from the output.

c. Archive the specific crash. Use the crash ID from the previous step:

```
$ oc exec -it $(oc get pod -n openshift-storage -l app=rook-ceph-operator -o name) -n openshift-storage -- ceph crash archive <crash-id> -c /var/lib/rook/openshift-storage/openshift-storage.config
```

Replace **crash-id** with the crash report.

d. Verify the crash has been archived:

```
$ oc exec -it $(oc get pod -n openshift-storage -l app=rook-ceph-operator -o name) -n openshift-storage -- ceph crash ls -c /var/lib/rook/openshift-storage/openshift-storage.config
```

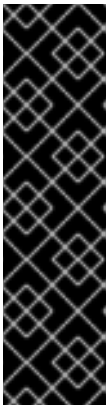
Verification

1. Verify that there is a new OSD running.

```
$ oc get -n openshift-storage pods -l app=rook-ceph-osd
```

Example output:

```
rook-ceph-osd-0-5f7f4747d4-snshw    1/1     Running     0          4m47s
rook-ceph-osd-1-85d99fb95f-2svc7    1/1     Running     0          1d20h
rook-ceph-osd-2-6c66cdb977-jp542    1/1     Running     0          1d20h
```



IMPORTANT

If the new OSD does not show as **Running** after a few minutes, restart the **rook-ceph-operator** pod to force a reconciliation.

```
$ oc delete pod -n openshift-storage -l app=rook-ceph-operator
```

Example output:

```
pod "rook-ceph-operator-6f74fb5bff-2d982" deleted
```

2. Verify that a new PVC is created.

```
$ oc get -n openshift-storage pvc | grep <lvs-name>
```

Example output:

```
ocs-deviceset-0-0-c2mqb   Bound   local-pv-b481410    1490Gi   RWO   localblock
5m
ocs-deviceset-1-0-959rp   Bound   local-pv-414755e0   1490Gi   RWO   localblock
1d20h
ocs-deviceset-2-0-79j94   Bound   local-pv-3e8964d3   1490Gi   RWO   localblock
1d20h
```

3. Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSDs are encrypted.

   a. Identify the nodes where the new OSD pods are running.

   ```
   $ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
   ```

   <OSD-pod-name>

   Is the name of the OSD pod.
   For example:

   ```
   $ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
   ```

   Example output:

   ```
   NODE
   compute-1
   ```

   b. For each of the nodes identified in the previous step, do the following:

      i. Create a debug pod and open a chroot environment for the selected host(s).

      ```
      $ oc debug node/<node name>
      ```

      <node name>

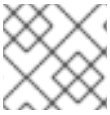
      Is the name of the node.

      ```
      $ chroot /host
      ```

      ii. Check for the **crypt** keyword beside the **ocs-deviceset** name(s).

      ```
      $ lsblk
      ```

4. Log in to OpenShift Web Console and check the OSD status on the storage dashboard.



NOTE

A full data recovery may take longer depending on the volume of data being recovered.

6.2. REPLACING STORAGE DEVICES DURING A HARDWARE REFRESH

When replacing a batch of smaller disks with larger disks as part of a hardware refresh, use the following procedure. This approach differs from single-device replacement because it manages rebalancing across multiple disks simultaneously, which reduces total migration time and cluster disruption.

Prerequisites

- Download the OpenShift Data Foundation command-line interface (CLI) tool. You can find a compatible version and download the CLI tool from the customer portal.
- Ensure the new replacement disks are available and accessible on the nodes.

Procedure

1. **Pause rebalancing (optional, recommended for batch replacements).**
   Use the OpenShift Data Foundation CLI to set **nobackfill** to prevent the cluster from rebalancing after each individual disk is added:

   ```
   $ odf ceph osd set nobackfill
   ```

2. **Add the new, larger capacity disks.**
   Use the OpenShift Web Console to add the replacement disks:

   a. Navigate to **Storage → Data Foundation → Storage Systems**.

   b. Click the kebab menu (⋮) next to your storage system and select **Add Capacity**.

   c. Select the appropriate Local Storage storage class and the number of new replacement disks.
   This creates the necessary persistent volumes (PVs) and persistent volume claims (PVCs) for the new OSDs automatically.

3. **Remove the old, smaller disks.**
   Use the **odf purge-osd** command to safely decommission each old OSD. This command checks whether it is safe to remove the OSD before draining and purging it:

   ```
   $ odf purge-osd <OSD_ID>
   ```

   Replace **<OSD_ID>** with the ID of the OSD to remove. The OSD ID is the integer in the pod name immediately after the **rook-ceph-osd** prefix. For example, in the pod name **rook-ceph-osd-0-6d77d6c7c6-m8xj6**, the OSD ID is **0**. Repeat this step for each old OSD being replaced.

   NOTE

   If removal of the OSD affects placement group (PG) status, you are prompted to enter **yes-force-destroy-osd** to confirm.

4. **Unset nobackfill and monitor recovery.**

After all new OSDs are active and all old OSDs are purged, re-enable rebalancing:

```
$ odf ceph osd unset nobackfill
```

Monitor the data resiliency status from the **Data Resiliency** card on the OpenShift Data Foundation dashboard in the OpenShift Web Console. The card displays a progress indicator for the ongoing rebalancing operation.

5. **(Optional) Adjust the recovery profile if migration is too slow.**
If the rebalancing rate is insufficient, use the OpenShift Data Foundation CLI to increase the recovery speed:

```
$ odf set recovery-profile high_recovery_ops
```



NOTE

Setting a high recovery profile increases recovery throughput but may affect cluster performance for running workloads. Reset the profile to the default when recovery is complete.

6.3. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON IBM POWER

You can replace an object storage device (OSD) in OpenShift Data Foundation deployed using local storage devices on IBM Power.



NOTE

There might be a need to replace one or more underlying storage devices.

Prerequisites

- Red Hat recommends that replacement devices are configured with similar infrastructure and resources to the device being replaced.
- Ensure that the data is resilient.
    - In the OpenShift Web Console, click **Storage → Data Foundation**.
    - Click the **Storage Systems** tab, and then click **ocs-storagecluster**.
    - In the **Status card** of **Block and File** dashboard, under the **Overview** tab, verify that *Data Resiliency* has a green tick mark.

Procedure

1. Identify the OSD that needs to be replaced and the OpenShift Container Platform node that has the OSD scheduled on it.

```
$ oc get -n openshift-storage pods -l app=rook-ceph-osd -o wide
```

Example output:

```
rook-ceph-osd-0-86bf8cdc8-4nb5t   0/1     crashLoopBackOff   0   24h   10.129.2.26
worker-0   <none>      <none>
rook-ceph-osd-1-7c99657cfb-jdzvz   1/1     Running   0         24h     10.128.2.46     worker-1
<none>      <none>
rook-ceph-osd-2-5f9f6dfb5b-2mnw9   1/1     Running   0         24h     10.131.0.33     worker-2
<none>      <none>
```

In this example, **rook-ceph-osd-0-86bf8cdc8-4nb5t** needs to be replaced and **worker-0** is the RHOCP node on which the OSD is scheduled.



NOTE

The status of the pod is **Running** if the OSD you want to replace is healthy.

2. Scale down the **rook-ceph-operator** and **ocs-operator** deployments.

```
$ oc -n openshift-storage scale deployment rook-ceph-operator --replicas=0
```

```
$ oc -n openshift-storage scale deployment ocs-operator --replicas=0
```

3. Scale down the OSD deployment for the OSD to be replaced.

```
$ osd_id_to_remove=0
```

```
$ oc scale -n openshift-storage deployment rook-ceph-osd-${osd_id_to_remove} --replicas=0
```

where, **osd_id_to_remove** is the integer in the pod name immediately after the **rook-ceph-osd** prefix. In this example, the deployment name is **rook-ceph-osd-0**.

Example output:

```
deployment.extensions/rook-ceph-osd-0 scaled
```

4. Verify that the **rook-ceph-osd** pod is terminated.

```
$ oc get -n openshift-storage pods -l ceph-osd-id=${osd_id_to_remove}
```

Example output:

```
No resources found in openshift-storage namespace.
```



IMPORTANT

If the **rook-ceph-osd** pod is in **terminating** state for more than a few minutes, use the **force** option to delete the pod.

```
$ oc delete -n openshift-storage pod rook-ceph-osd-0-86bf8cdc8-4nb5t --grace-period=0 --force
```

Example output:

```
warning: Immediate deletion does not wait for confirmation that the running resource has been terminated. The resource may continue to run on the cluster indefinitely.
  pod "rook-ceph-osd-0-86bf8cdc8-4nb5t" force deleted
```

5. Remove the old OSD from the cluster so that you can add a new OSD.

   a. Identify the **DeviceSet** associated with the OSD to be replaced.

   ```
   $ oc get -n openshift-storage -o yaml deployment rook-ceph-osd-${osd_id_to_remove} | grep ceph.rook.io/pvc
   ```

   Example output:

   ```
   ceph.rook.io/pvc: ocs-deviceset-localblock-0-data-0-64xjl
     ceph.rook.io/pvc: ocs-deviceset-localblock-0-data-0-64xjl
   ```

   In this example, the Persistent Volume Claim (PVC) name is **ocs-deviceset-localblock-0-data-0-64xjl**.

   b. Identify the Persistent Volume (PV) associated with the PVC.

   ```
   $ oc get -n openshift-storage pvc ocs-deviceset-<x>-<y>-<pvc-suffix>
   ```

   where, ***x***, ***y***, and ***pvc-suffix*** are the values in the **DeviceSet** identified in an earlier step.

   Example output:

   ```
   NAME                                  STATUS   VOLUME        CAPACITY   ACCESS MODES   STORAGECLASS   AGE
   ocs-deviceset-localblock-0-data-0-64xjl   Bound   local-pv-8137c873   256Gi   RWO   localblock   24h
   ```

   In this example, the associated PV is **local-pv-8137c873**.

   c. Identify the name of the device to be replaced.

   ```
   $ oc get pv local-pv-<pv-suffix> -o yaml | grep path
   ```

   where, ***pv-suffix*** is the value in the PV name identified in an earlier step.

   Example output:

```
path: /mnt/local-storage/localblock/vdc
```

In this example, the device name is **vdc**.

d. Identify the **prepare-pod** associated with the OSD to be replaced.

```
$ oc describe -n openshift-storage pvc ocs-deviceset-<x>-<y>-<pvc-suffix> | grep Used
```

where, ***x***, ***y***, and ***pvc-suffix*** are the values in the **DeviceSet** identified in an earlier step.

Example output:

```
Used By:    rook-ceph-osd-prepare-ocs-deviceset-localblock-0-data-0-64knzkc
```

In this example, the **prepare-pod** name is **rook-ceph-osd-prepare-ocs-deviceset-localblock-0-data-0-64knzkc**.

e. Delete any old **ocs-osd-removal** jobs.

```
$ oc delete -n openshift-storage job ocs-osd-removal-job
```

Example output:

```
job.batch "ocs-osd-removal-job" deleted
```



NOTE

The above command must reach **Completed** state before moving to the next steps. This can take more than ten minutes.

f. Change to the **openshift-storage** project.

```
$ oc project openshift-storage
```

g. Remove the old OSD from the cluster.

```
$ oc process -n openshift-storage ocs-osd-removal -p
FAILED_OSD_IDS=${osd_id_to_remove} FORCE_OSD_REMOVAL=false |oc create -n
openshift-storage -f -
```

Set **FORCE_OSD_REMOVAL** to **true** when removing an OSD leaves the cluster with insufficient OSDs to satisfy its data protection requirements. Without this parameter, the Ceph safe-to-destroy check does not pass and the removal job hangs indefinitely.



WARNING

Proceeding will permanently remove the OSD from the cluster and destroy its data. Ensure **osd_id_to_remove** is correct. Do not process multiple IDs from the same failure domain at once. Contact Red Hat Technical Support if you need assistance.

6. Verify that the OSD was removed successfully by checking the status of the **ocs-osd-removal-job** pod.
   A status of **Completed** confirms that the OSD removal job succeeded.

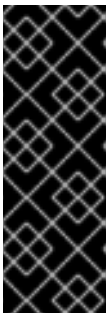
```
$ oc get pod -l job-name=ocs-osd-removal-job -n openshift-storage
```

7. Ensure that the OSD removal is completed.

```
$ oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1 | egrep -i 'completed removal'
```

   Example output:

```
2022-05-10 06:50:04.501511 I | cephosd: completed removal of OSD 0
```



IMPORTANT

If the **ocs-osd-removal-job** fails and the pod is not in the expected **Completed** state, check the pod logs for further debugging.

For example:

```
# oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1
```

8. If encryption was enabled at the time of install, remove **dm-crypt** managed **device-mapper** mapping from the OSDs that are removed from the respective OpenShift Data Foundation nodes.

   a. Get the PVC name(s) of the replaced OSD(s) from the logs of **ocs-osd-removal-job** pod.

```
$ oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1  |egrep -i ‘pvc|deviceset’
```

      Example output:

```
2021-05-12 14:31:34.666000 I | cephosd: removing the OSD PVC "ocs-deviceset-xxxx-xxx-xxx-xxx"
```

   b. For each of the previously identified nodes, do the following:

i. Create a **debug** pod and **chroot** to the host on the storage node.

```
$ oc debug node/<node name>
```

<node name>

Is the name of the node.

```
$ chroot /host
```

ii. Find the relevant device name based on the PVC names identified in the previous step.

```
$ dmsetup ls| grep <pvc name>
```

<pvc name>

Is the name of the PVC.
Example output:

```
ocs-deviceset-xxx-xxx-xxx-xxx-block-dmcrypt (253:0)
```

iii. Remove the mapped device.

```
$ cryptsetup luksClose --debug --verbose ocs-deviceset-xxx-xxx-xxx-xxx-block-dmcrypt
```



IMPORTANT

If the above command gets stuck due to insufficient privileges, run the following commands:

- Press **CTRL+Z** to exit the above command.
- Find the PID of the process which was stuck.

```
$ ps -ef | grep crypt
```

- Terminate the process using the **kill** command.

```
$ kill -9 <PID>
```

<PID>

Is the process ID.

- Verify that the device name is removed.

```
$ dmsetup ls
```

9. Find the PV that need to be deleted.

```
$ oc get pv -L kubernetes.io/hostname | grep localblock | grep Released
```

Example output:

```
local-pv-d6bf175b 1490Gi RWO Delete Released openshift-storage/ocs-deviceset-0-data-0-6c5pw localblock 2d22h compute-1
```

10. Delete the PV.

```
$ oc delete pv <pv-name>
```

<pv-name>

Is the name of the PV.

11. Replace the old device and use the new device to create a new OpenShift Container Platform PV.

    a. Log in to the OpenShift Container Platform node with the device to be replaced. In this example, the OpenShift Container Platform node is **worker-0**.

    ```
    $ oc debug node/worker-0
    ```

    Example output:

    ```
    Starting pod/worker-0-debug ...
    To use host binaries, run `chroot /host`
    Pod IP: 192.168.88.21
    If you don't see a command prompt, try pressing enter.
    # chroot /host
    ```

    b. Record the **/dev/disk** that is to be replaced using the device name, **vdc**, identified earlier.

    ```
    # ls -alh /mnt/local-storage/localblock
    ```

    Example output:

    ```
    total 0
    drwxr-xr-x. 2 root root 17 Nov 18 15:23 .
    drwxr-xr-x. 3 root root 24 Nov 18 15:23 ..
    lrwxrwxrwx. 1 root root 8 Nov 18 15:23 vdc -> /dev/vdc
    ```

    c. Find the name of the **LocalVolume** CR, and remove or comment out the device **/dev/disk** that is to be replaced.

    ```
    $ oc get -n openshift-local-storage localvolume
    ```

    Example output:

    ```
    NAME AGE
    localblock 25h
    ```

    ```
    # oc edit -n openshift-local-storage localvolume localblock
    ```

    Example output:

```
[...]
    storageClassDevices:
    - devicePaths:
   #   - /dev/vdc
      storageClassName: localblock
      volumeMode: Block
[...]
```

Make sure to save the changes after editing the CR.

12. Log in to the OpenShift Container Platform node with the device to be replaced and remove the old **symlink**.

```
$ oc debug node/worker-0
```

Example output:

```
Starting pod/worker-0-debug ...
To use host binaries, run `chroot /host`
Pod IP: 192.168.88.21
If you don't see a command prompt, try pressing enter.
# chroot /host
```

a. Identify the old **symlink** for the device name to be replaced. In this example, the device name is **vdc**.

```
# ls -alh /mnt/local-storage/localblock
```

Example output:

```
total 0
drwxr-xr-x. 2 root root 17 Nov  18 15:23 .
drwxr-xr-x. 3 root root 24 Nov  18 15:23 ..
lrwxrwxrwx. 1 root root  8 Nov  18 15:23 vdc -> /dev/vdc
```

b. Remove the **symlink**.

```
# rm /mnt/local-storage/localblock/vdc
```

c. Verify that the **symlink** is removed.

```
# ls -alh /mnt/local-storage/localblock
```

Example output:

```
total 0
drwxr-xr-x. 2 root root 6 Nov 18 17:11 .
drwxr-xr-x. 3 root root 24 Nov 18 15:23 ..
```

13. Replace the old device with the new device.

14. Log back into the correct OpenShift Container Platform node and identify the device name for the new drive. The device name must change unless you are resetting the same device.

```
# lsblk
```

Example output:

```
NAME                         MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
vda                          252:0    0   40G  0 disk
|-vda1                       252:1    0    4M  0 part
|-vda2                       252:2    0  384M  0 part /boot
`-vda4                       252:4    0 39.6G  0 part
  `-coreos-luks-root-nocrypt 253:0    0 39.6G  0 dm   /sysroot
vdb                          252:16   0  512B  1 disk
vdd                          252:32   0  256G  0 disk
```

In this example, the new device name is **vdd**.

15. After the new **/dev/disk** is available, you can add a new disk entry to the LocalVolume CR.

    a. Edit the LocalVolume CR and add the new **/dev/disk**.
    In this example, the new device is **/dev/vdd**.

    ```
    # oc edit -n openshift-local-storage localvolume localblock
    ```

    Example output:

    ```
    [...]
      storageClassDevices:
      - devicePaths:
      #  - /dev/vdc
        - /dev/vdd
        storageClassName: localblock
        volumeMode: Block
    [...]
    ```

    Make sure to save the changes after editing the CR.

16. Verify that there is a new PV in **Available** state and of the correct size.

    ```
    $ oc get pv | grep 256Gi
    ```

    Example output:

    ```
    local-pv-1e31f771   256Gi   RWO   Delete  Bound  openshift-storage/ocs-deviceset-
    localblock-2-data-0-6xhkf   localblock   24h
    local-pv-ec7f2b80   256Gi   RWO   Delete  Bound  openshift-storage/ocs-deviceset-
    localblock-1-data-0-hr2fx   localblock   24h
    local-pv-8137c873   256Gi   RWO   Delete  Available
    localblock   32m
    ```

17. Create a new OSD for the new device.
    Deploy the new OSD. You need to restart the **rook-ceph-operator** to force operator reconciliation.

    a. Identify the name of the **rook-ceph-operator**.

    ```
    -
    ```

```
$ oc get -n openshift-storage pod -l app=rook-ceph-operator
```

Example output:

```
NAME                                  READY   STATUS    RESTARTS   AGE
rook-ceph-operator-85f6494db4-sg62v   1/1     Running   0          1d20h
```

b. Delete the **rook-ceph-operator**.

```
$ oc delete -n openshift-storage pod rook-ceph-operator-85f6494db4-sg62v
```

Example output:

```
pod "rook-ceph-operator-85f6494db4-sg62v" deleted
```

In this example, the rook-ceph-operator pod name is **rook-ceph-operator-85f6494db4-sg62v**.

c. Verify that the **rook-ceph-operator** pod is restarted.

```
$ oc get -n openshift-storage pod -l app=rook-ceph-operator
```

Example output:

```
NAME                                  READY   STATUS    RESTARTS   AGE
rook-ceph-operator-85f6494db4-wx9xx   1/1     Running   0          50s
```

Creation of the new OSD may take several minutes after the operator restarts.

18. Delete the **ocs-osd-removal** job(s).

```
$ oc delete -n openshift-storage job ocs-osd-removal-job
```

Example output:

```
job.batch "ocs-osd-removal-job" deleted
```



NOTE

When using an external key management system (KMS) with data encryption, the old OSD encryption key can be removed from the Vault server as it is now an orphan key.

19. Scale up the **rook-ceph-operator** and **ocs-operator** deployments.

```
$ oc -n openshift-storage scale deployment rook-ceph-operator --replicas=1
```

```
$ oc -n openshift-storage scale deployment ocs-operator --replicas=1
```

Verification

1. Verify that there is a new OSD running.

```
$ oc get -n openshift-storage pods -l app=rook-ceph-osd
```

Example output:

```
rook-ceph-osd-0-76d8fb97f9-mn8qz   1/1     Running   0          23m
rook-ceph-osd-1-7c99657cfb-jdzvz   1/1     Running   1          25h
rook-ceph-osd-2-5f9f6dfb5b-2mnw9   1/1     Running   0          25h
```

2. Verify that a new PVC is created.

```
$ oc get -n openshift-storage pvc | grep localblock
```

Example output:

```
ocs-deviceset-localblock-0-data-0-q4q6b   Bound    local-pv-8137c873       256Gi      RWO
localblock         10m
ocs-deviceset-localblock-1-data-0-hr2fx   Bound    local-pv-ec7f2b80       256Gi      RWO
localblock         1d20h
ocs-deviceset-localblock-2-data-0-6xhkf   Bound    local-pv-1e31f771       256Gi      RWO
localblock         1d20h
```

3. Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.

   a. Identify the nodes where the new OSD pods are running.

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
```

<OSD-pod-name>

Is the name of the OSD pod.
For example:

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
```

Example output:

```
NODE
compute-1
```

   b. For each of the previously identified nodes, do the following:

      i. Create a debug pod and open a chroot environment for the selected host(s).

```
$ oc debug node/<node name>
```

<node name>

Is the name of the node.

-

```
$ chroot /host
```

ii. Check for the **crypt** keyword beside the **ocs-deviceset** name(s).

```
$ lsblk
```

4. Log in to OpenShift Web Console and check the status card in the OpenShift Data Foundation dashboard under Storage section.



NOTE

A full data recovery may take longer depending on the volume of data being recovered.

CHAPTER 7. REMOVING OSDS USING THE OPENSHIFT DATA FOUNDATION CLI TOOL

7.1. REMOVING OBJECT STORAGE DEVICES USING THE OPENSHIFT DATA FOUNDATION CLI TOOL

You can use the OpenShift Data Foundation command line interface (CLI) tool to automate the process of object storage device (OSD) removal. This helps to avoid the possible data loss while removing OSDs.

Prerequisites

- Download the OpenShift Data Foundation command line interface (CLI) tool. With the Data Foundation CLI tool, you can effectively manage and troubleshoot your Data Foundation environment from a terminal. You can find a compatible version and download the CLI tool from the customer portal.

Procedure

1. Identify the OSD that needs to be removed. Use this tool only if the OSD is in **CrashLoopBackOff** or **Error** state.

   ```
   $ oc get -n openshift-storage pods -l app=rook-ceph-osd -o wide
   ```

   Example output:

   ```
   rook-ceph-osd-0-6d77d6c7c6-m8xj6   0/1   CrashLoopBackOff   0   24h   10.129.0.16
   compute-2   <none>          <none>
   ```

2. Run the following command to remove OSD **0**:

   ```
   $ odf purge-osd 0
   ```

3. [Optional] If removal of the OSD affects placement group (PG) status, enter **yes-force-destroy-osd**.

4. Verify that the last line of the command output contains **cephosd: completed removal of OSD 0**.

5. Verify the corresponding deployment is removed:

   ```
   $ oc get deployment rook-ceph-osd-0
   ```